

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 3: Encoding & Transmission Modes

Course: Computer Networks (Credit Hours: 3)

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1. WHAT IS ENCODING? (SPEAKING THE CABLE'S LANGUAGE)

Inside a computer, data is stored as **0s and 1s**. But a copper wire or optical fiber cannot read numbers directly; it only understands electrical voltages or light flashes.

Definition: Encoding is the process of converting digital data (0s and 1s) into electrical signals or light pulses that can travel across a transmission wire without error.

Easy Example Translating English text into Morse code (dots and dashes) so it can be transmitted by flashlight.

2. THREE SIMPLE LINE CODING SCHEMES

Encoding Scheme	How It Works (Easy Rule)	Real-World Use
1. Unipolar (NRZ)	Uses only 1 polarity: High voltage (+5V) = 1 , Zero voltage (0V) = 0 . Very simple, but wastes power.	Inside simple internal circuits.
2. Polar NRZ-L	Uses two opposite voltages: Positive voltage (+V) = 0 , Negative voltage (-V) = 1 .	Basic serial port communications.
3. Manchester Encoding	The signal transitions in the middle of every bit : High-to-Low = 0 , Low-to-High = 1 . Solves synchronization issues!	Classic 10 Mbps Ethernet (LAN cables).

3. SYNCHRONOUS VS. ASYNCHRONOUS TRANSMISSION

When two computers communicate, how do they keep their internal clocks in sync so they know when a bit starts and ends?

A. Asynchronous Transmission

Data is sent **one character (1 byte) at a time**. Sender and receiver do not share a synchronized clock.

- Uses a **Start Bit (0)** at the beginning and a **Stop Bit (1)** at the end of each byte.
- Gaps between characters can be random.
- **Easy Example** Typing on a keyboard: letters arrive only when you press a key, with irregular pauses.

B. Synchronous Transmission

Data is sent in **large continuous blocks (frames)** without individual start and stop bits.

- Sender and receiver share a synchronized clock signal.
- No gaps between bytes; much faster and has very low overhead.
- **Easy Example** Marching soldiers stepping together in exact lockstep rhythm. High-speed file downloads over LAN.

4. SUMMARY COMPARISON: ASYNCHRONOUS VS. SYNCHRONOUS

Feature	Asynchronous	Synchronous
Data Unit	1 byte (character by character)	Full blocks or frames of data
Start & Stop Bits	Yes (required for each byte)	No (uses clock timing and sync flags)
Speed	Slower (extra framing overhead)	Much faster and highly efficient
Typical Use	Keyboard, mouse, legacy serial links	High-speed Ethernet, fiber backbones

5. QUICK STUDENT REVIEW QUESTIONS

Q1: Why do we add Start and Stop bits in Asynchronous transmission?

Ans: Because the sender and receiver do not share a common clock. The start bit warns the receiver "incoming byte!", and the stop bit signals "byte finished!"

Q2: Why is Manchester Encoding favored in Ethernet LANs?

Ans: Because it changes voltage in the middle of every bit, allowing the receiving device to lock onto the sender's clock automatically without losing track.